



UNITED STATES
NUCLEAR REGULATORY COMMISSION
WASHINGTON, D.C. 20555-0001

July 8, 2024

Mr. Jeongsun Kim
Quality Assurance General Manager
Doosan Enerbility, Ltd.
22, DoosanVolvo-ro, Seongsan-gu, Changwon-si
Gyeongnam, 51711, Republic of Korea

SUBJECT: NUCLEAR REGULATORY COMMISSION VENDOR INSPECTION REPORT OF
DOOSAN ENERBILITY, LTD. NO. 99901373/2024-201, AND NOTICE OF
VIOLATION

Dear Mr. Kim:

On May 20 - 24, 2024, the U.S. Nuclear Regulatory Commission (NRC) staff conducted an inspection at the Doosan Enerbility, Ltd.'s facility (hereafter referred to as Doosan) in Changwon, Korea. The purpose of this limited-scope routine inspection was to assess Doosan's compliance with the provisions of Title 10 of the *Code of Federal Regulations* (10 CFR) Part 21, "Reporting of Defects and Noncompliance," and selected portions of Appendix B, "Quality Assurance Program Criteria for Nuclear Power Plants and Fuel Reprocessing Plants," to 10 CFR Part 50, "Domestic Licensing of Production and Utilization Facilities," and of Subpart G, "Quality Assurance," of 10 CFR Part 72, "Licensing Requirements for the Independent Storage of Spent Nuclear Fuel, High-Level Radioactive Waste, and Reactor-Related Greater Than Class C Waste."

This technically-focused inspection specifically evaluated Doosan's implementation of quality activities associated with the construction of safety-related and important to safety components for U.S. nuclear power plants. The enclosed report presents the results of the inspection. This NRC inspection report does not constitute NRC's endorsement of Doosan's overall quality assurance (QA) or 10 CFR Part 21 programs.

Based on the results of this inspection, the NRC has determined that one Severity Level IV violation of NRC requirements occurred. The violation is cited in the enclosed Notice of Violation (NOV) and the circumstances surrounding it are described in detail in the subject inspection report. The violation is being cited in the NOV because Doosan failed to specify that the provisions of 10 CFR Part 21 apply in the procurement documents for safety-related and important to safety components in accordance with the requirements of 10 CFR 21.31, "Procurement documents."

You are required to respond to this letter and should follow the instructions specified in the enclosed NOV when preparing your response. If you have additional information that you believe the NRC should consider, you may provide it in your response to the NOV. The NRC will use your response, in part, to determine whether further enforcement action is necessary to ensure compliance with regulatory requirements.

In accordance with 10 CFR 2.390, "Public inspections, exemptions, requests for withholding," of the NRC's "Rules of Practice," a copy of this letter, its enclosure(s), and your response will be made available electronically for public inspection in the NRC's Public Document Room or from the NRC's document system (ADAMS), accessible at <http://www.nrc.gov/readingrm/adams.html>. To the extent possible, your response (if applicable), should not include any personal privacy, proprietary, or safeguards information so that it can be made available to the public without redaction. If personal privacy or proprietary information is necessary to provide an acceptable response, please provide a bracketed copy of your response that identifies the information that should be protected and a redacted copy of your response that deletes such information. If you request that such material be withheld from public disclosure, you must specifically identify the portions of your response that you seek to have withheld and provide in detail the bases for your claim (e.g., explain why the disclosure of information would create an unwarranted invasion of personal privacy or provide the information required by 10 CFR 2.390(b) to support a request for withholding confidential commercial or financial information).

If safeguards information is necessary to provide an acceptable response, please provide the level of protection described in 10 CFR 73.21, "Protection of Safeguards Information: Performance Requirements."

Sincerely,



Signed by Kavanagh, Kerri
on 07/08/24

Kerri A. Kavanagh, Chief
Quality Assurance and Vendor Inspection Branch
Division of Reactor Oversight
Office of Nuclear Reactor Regulation

Docket No.: 99901373

EPID No.: I-2024-201-0027

Enclosures:

1. Notice of Violation
2. Inspection Report No. 99901373/2024-201
and Attachment

SUBJECT: NUCLEAR REGULATORY COMMISSION VENDOR INSPECTION REPORT OF
DOOSAN ENERBILITY, LTD. NO. 9990137/2024-201, AND NOTICE OF
VIOLATION DATE: July 8, 2024

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NRR-106

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NAME	BHughes	KKavanagh	
DATE	6/26/2024	7/8/2024	

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NOTICE OF VIOLATION

Doosan Enerbility, Ltd.
22, DoosanVolvo-ro, Seongsan-gu, Changwon-si
Gyeongnam, 51711, Republic of Korea

Docket No. 99901373
Report No. 2024-201

During a U.S. Nuclear Regulatory Commission (NRC) inspection conducted at the Doosan Enerbility, Ltd.'s (hereafter referred to as Doosan) facility in Changwon, Korea, on May 20, 2024 through May 24, 2024, a violation of NRC requirements was identified. In accordance with the NRC's Enforcement Policy, the violation is listed below:

Section 21.31, "Procurement documents," of Title 10 of the *Code of Federal Regulations* (10 CFR) Part 21, requires that "Each individual, corporation, partnership, dedicating entity, or other entity subject to the regulations in this part shall ensure that each procurement document for a facility, or a basic component issued by him, her or it on or after January 6, 1978, specifies, when applicable, that the provisions of 10 CFR Part 21 apply."

Step 5.9 in Section 5.0, "General," of Doosan's Quality Assurance Procedure No. PQAP-1602, "Reporting of 'Defects' and 'Failures' to Comply Pursuant to 10CFR21," Revision 13, dated May 10, 2024, states that "Doosan's procurement document shall specify the applicability of 10 CFR 21."

Contrary to the above, as of May 24, 2024, Doosan failed to specify in procurement documents for suppliers of safety-related services and important to safety materials to be procured as basic components that the provisions of 10 CFR Part 21 apply. Specifically, for the procurement of safety-related machining services, Doosan invoked in the purchase orders (POs) that suppliers meet purchase specification No. CF-MPS-001, "General Requirements (for foreign company)," Revision 17, dated January 20, 2023, rather than imposing the applicable requirements of 10 CFR Part 21. Doosan developed CF-MPS-001 to address the fact that Doosan's suppliers of machining services do not implement a program that meets the requirements of 10 CFR Part 21. This procedure relieves Doosan's machining suppliers from their responsibilities under 10 CFR Part 21 and requires the suppliers to notify Doosan of any nonconformances identified in the machining services supplied to Doosan. In addition, for the procurement of important to safety materials for a vertical concrete cask, Doosan did not invoke the requirements of 10 CFR Part 21 in the POs.

This issue has been identified as Violation 99901373/2024-201-01.

This is a Severity Level IV violation (Section 6.9.d of the NRC Enforcement Policy).

Pursuant to the provisions of 10 CFR Part 2.201, "Notice of Violation," Doosan is hereby required to submit a written statement or explanation to the U.S. Nuclear Regulatory Commission, ATTN: Document Control Desk, Washington, DC 20555-0001, with a copy to the Chief, Quality Assurance and Vendor Inspection Branch, Division of Reactor Oversight, Office of Nuclear Reactor Regulation, within 30 days of the date of the letter transmitting this Notice of Violation. This reply should be clearly marked as a "Reply to a Notice of Violation" and should include for each violation: (1) the reason for the violation or, if contested, the basis for disputing the violation or severity level, (2) the corrective steps that have been taken and the results achieved, (3) the corrective steps that will be taken to avoid further violations, and (4) the date

when full compliance will be achieved. Your response may reference or include previous docketed correspondence if the correspondence adequately addresses the required response. Where good cause is shown, consideration will be given to extending the response time.

If you contest this enforcement action, provide a copy of your response, with the basis for your denial, to the Director, Office of Enforcement, United States Nuclear Regulatory Commission, Washington, DC 20555-0001.

Because your response will be made available electronically for public inspection in the NRC's Public Document Room or from the NRC's document system (ADAMS), accessible from the NRC's Web site at <http://www.nrc.gov/reading-rm/adams.html>, to the extent possible, it should not include any personal privacy, proprietary, or safeguards information so that it can be made available to the public without redaction. If personal privacy or proprietary information is necessary to provide an acceptable response, then please provide a bracketed copy of your response that identifies the information that should be protected and a redacted copy of your response that deletes such information.

If you request that such material be withheld, you must specifically identify the portions of your response that you seek to have withheld and provide in detail the bases for your claim of withholding (e.g., explain why the disclosure of information would create an unwarranted invasion of personal privacy or provide the information required by 10 CFR 2.390(b) to support a request for withholding confidential commercial or financial information). If safeguards information is necessary to provide an acceptable response, please provide the level of protection described in 10 CFR 73.21, "Protection of Safeguards Information: Performance Requirements."

Dated this 8th day of July 2024.

**U.S. NUCLEAR REGULATORY COMMISSION
OFFICE OF NUCLEAR REACTOR REGULATION
DIVISION OF REACTOR OVERSIGHT
VENDOR INSPECTION REPORT**

Docket No.: 99901373

Report No.: 99901373/2024-201

Vendor: Doosan Enerbility Company, Ltd.
22, DoosanVolvo-ro, Seongsan-gu, Changwon-si
Gyeongnam, 51711, Republic of Korea

Vendor Contact: Mr. Kyeongseon Lim
Quality Assurance Senior Engineer
Email: Kyeongseon.lim@doosan.com
Phone: (82) 55-278-3305

Nuclear Industry Activity: Doosan Enerbility, Ltd. (hereafter referred to as Doosan) is one of the largest manufacturers of reactor vessels, steam generators, reactor internals, and balance of plant components for the worldwide nuclear industry. Doosan provides components to the U.S. nuclear power reactors as replacement components and for the AP1000 units recently constructed. Doosan was selected by Westinghouse Electric Company to manufacture the lower assembly forgings of the replacement steam generators for the Surry Power Station. Doosan was also selected by NAC International to manufacture important to safety materials related to a transportable canister for NAC International's MAGANASTOR spent fuel storage system, and a transfer cask for use at Three Mile Island as part of decommissioning.

Inspection Dates: May 20 - 24, 2024

Inspectors: Yamir Diaz-Castillo NRR/DRO/IQVB Team Leader
John Honcharik NRR/DNRL/NPHP Technical Specialist
Marlone Davis NMSS/DFM/IOB Sr. Safety Inspector

Approved by: Kerri A. Kavanagh, Chief
Quality Assurance and Vendor Inspection Branch
Division of Reactor Oversight
Office of Nuclear Reactor Regulation

EXECUTIVE SUMMARY

Doosan Enerbility, Ltd., LLC
99901373/2024-201

The U.S. Nuclear Regulatory Commission (NRC) staff conducted a limited-scope routine vendor inspection at the Doosan Enerbility, Ltd.'s (hereafter referred to as Doosan) facility in Changwon, Korea, to verify that it had implemented an adequate quality assurance (QA) program that complies with the requirements of Appendix B, "Quality Assurance Criteria for Nuclear Power Plants and Fuel Reprocessing Plants," to Title 10 of the *Code of Federal Regulations* (10 CFR) Part 50, "Domestic Licensing of Production and Utilization Facilities," and Subpart G, "Quality Assurance," of 10 CFR Part 72, "Licensing Requirements for the Independent Storage of Spent Nuclear Fuel, High-Level Radioactive Waste, and Reactor-Related Greater Than Class C Waste." In addition, the NRC inspection verified that Doosan implemented a program under 10 CFR Part 21, "Reporting of Defects and Noncompliance." Furthermore, the NRC inspection verified that Doosan had implemented a program in accordance with the applicable requirements of Section III, "Rules for Construction of Nuclear Facility Components," Section V, "Nondestructive Examination," and Section IX, "Welding and Brazing Qualification," of the American Society of Mechanical Engineers (ASME) Boiler and Pressure Vessel (B&PV) Code, and the American Society for Nondestructive Testing (ASNT) SNT-TC-1A, "Personnel Qualification and Certification in Nondestructive Testing." This was the 5th NRC inspection of Doosan's facility in Changwon, Korea. The last NRC inspection of Doosan occurred in July 2014.

This technically-focused inspection specifically evaluated Doosan's implementation of quality activities associated with the construction of safety-related and important to safety components for U.S. nuclear power plants. Specific activities observed by the NRC inspection team included:

- Post weld heat treatment of lifting lugs on tubesheet No. F23189-010 for the replacement steam generators (RSGs) for Surry Power Station Units 1 and 2 performed in accordance with heat treat work order No. 238-300 and heat treatment procedure, PTC-NSSS-001, "Heat Treatment Work Procedure Surry 1 & 2," Revision 0, as documented in traveler No. F23189060-2.
- Ultrasonic non-destructive examination (NDE) on tubesheet forging No. F231890-030 for the RSGs for Surry Power Station Units 1 and 2 performed in accordance Process Specification No. FN231890, "Process Specification for NDE Procedure for Surry 1 & 2 Tubesheet," Revision 7, as documented in traveler No. F23189030.
- Temporary storage rework for the Watts Bar Nuclear Plant Unit 2 Reactor Closure Vessel Head in accordance with traveler No. 101920517 based on revised purchase order No. 4500634344.
- Temporary storage of two transportable storage canisters (TSCs) for the NAC MAGNASTOR dry cask storage system. This included temporary packing of completed fabricated important to safety components, which also included two TSC closure lids and rings related to the canister.

These regulations served as the bases for the NRC inspection:

- Appendix B to 10 CFR Part 50
- 10 CFR Part 21
- Subpart G to 10 CFR Part 72

During this inspection, the NRC inspection team implemented Inspection Procedure (IP) 43002, "Routine Inspections of Nuclear Vendors," dated February 10, 2023; IP 43004, "Inspection of Commercial-Grade Dedication Programs," dated February 10, 2023; IP 36100, "Inspection of 10 CFR Part 21 and Programs for Reporting Defects and Noncompliance," dated February 10, 2023, and IP 60852, "ISFSI Component Fabrication by Outside Fabricators," dated August 17, 2023.

With the exception of the violation described below, the NRC inspection team concluded that Doosan's QA policies and procedures comply with the applicable requirements of Appendix B to 10 CFR Part 50, 10 CFR Part 21, and Subpart G to 10 CFR Part 72, and that Doosan's personnel are implementing these policies and procedures effectively. The results of this inspection are summarized below.

10 CFR Part 21 Program

The NRC inspection team issued Violation 99901373/2024-201-01 in association with Doosan's failure to implement the regulatory requirements of 10 CFR Part 21. Violation 99901373/2024-201-01 cites Doosan for failing to specify in procurement documents for suppliers of safety-related services and important to safety materials to be procured as basic components that the provisions of 10 CFR Part 21 apply. Doosan initiated Corrective Action Report No. 2440053 to address this issue.

Other Inspection Areas

The NRC inspection team determined that Doosan established its programs for design control, procurement document control, supplier oversight, material traceability, control of special processes, test control, control of measuring and test equipment, nonconformances, corrective action, and internal audits in accordance with the applicable regulatory requirements of Appendix B to 10 CFR Part 50 and Subpart G to 10 CFR Part 72, as applicable. Based on the limited sample of documents reviewed and activities observed, the NRC inspection team also determined that Doosan is implementing its policies and procedures associated with these programs. No findings of significance were identified in these areas.

REPORT DETAILS

1. 10 CFR Part 21 Program

a. Inspection Scope

The NRC inspection team reviewed Doosan Enerbility, Ltd.'s (hereafter referred to as Doosan) policies and implementing procedures that govern the implementation of its Title 10 of the *Code of Federal Regulations* (10 CFR) Part 21, "Reporting of Defects and Noncompliance," program to verify compliance with the regulatory requirements. The NRC inspection team evaluated the 10 CFR Part 21 postings and a sample of Doosan's purchase orders (POs) for compliance with the requirements of 10 CFR 21.6, "Posting requirements," and 10 CFR 21.31, "Procurement documents," respectively. The NRC inspection team also verified that Doosan's nonconformance and corrective action procedures provide a link to the 10 CFR Part 21 program.

The NRC inspection team also discussed the 10 CFR Part 21 program with Doosan's management and technical staff. The attachment to this inspection report lists the documents reviewed and personnel interviewed by the NRC inspection team.

b. Observations and Findings

During the review of a sample of POs for safety-related machining services and important to safety materials, the NRC inspection team noted that the POs did not specify that the provisions of 10 CFR Part 21 apply. For the procurement of safety-related machining services, rather than imposing that the requirements of 10 CFR Part 21 apply, Doosan imposed in the POs purchase specification No. CF-MPS-001, "General Requirements (for foreign company)," Revision 17, dated January 20, 2023. Doosan developed CF-MPS-001 to address the fact that Doosan's chosen suppliers of machining services do not implement a program that meets the requirements of 10 CFR Part 21. This procedure relieved Doosan's machining suppliers from their responsibilities under 10 CFR Part 21 and required the suppliers to notify Doosan of any nonconformances identified in the machining services supplied to Doosan. The regulations in 10 CFR Part 21 do not allow for suppliers of basic components to relieve their sub-suppliers of the responsibilities under 10 CFR Part 21 when procuring services as basic components. In addition, for the procurement of important to safety materials for a vertical concrete cask, Doosan did not invoke the requirements of 10 CFR Part 21 in the POs.

The NRC inspection team identified this issue as Violation 99901373/2024-201-01 for Doosan's failure to specify in the procurement documents for suppliers of safety-related services and important to safety materials to be procured as basic components that the provisions of 10 CFR Part 21 apply. Doosan initiated Corrective Action Report (CAR) No. 2440053 to address this issue.

c. Conclusion

The NRC inspection team issued Violation 99901373/2024-201-01 in association with Doosan's failure to implement the regulatory requirements of 10 CFR Part 21. Violation 99901373/2024-201-01 cites Doosan for failing to specify in the procurement documents

for suppliers of safety-related services and important to safety materials to be procured as basic components that the provisions of 10 CFR Part 21 apply.

2. Design Control

a. Inspection Scope

The NRC inspection team reviewed Doosan's policies and implementing procedures that govern the implementation of its design control program to verify compliance with the regulatory requirements of Criterion III, "Design Control," of Appendix B, "Quality Assurance Criteria for Nuclear Power Plants and Fuel Reprocessing Plants," to 10 CFR Part 50, "Domestic Licensing of Production and Utilization Facilities," and 10 CFR 72.146, "Design control," of Subpart G, "Quality Assurance," to 10 CFR Part 72, "Licensing Requirements for the Independent Storage of Spent Nuclear Fuel, High-Level Radioactive Waste, and Reactor-Related Greater Than Class C Waste."

The NRC inspection team confirmed that Doosan's design control process provided provisions for the control of design inputs, outputs, design analyses, records, and organizational interfaces, as applicable. The NRC inspection team reviewed a sample of PO's, purchase specifications, and fabrication drawings associated with the design of the replacement steam generators (RSG) for the Surry Power Station Units 1 and 2, and for vertical concrete cask liners for the NAC MAGNASTOR dry cask storage system. The NRC inspection team reviewed the process for translating the design inputs and design specifications into manufacturing procedures and fabrication drawings. The NRC inspection team also reviewed Doosan's design control processes to verify that procedures included the control of design changes and that those design changes are subject to design control measures commensurate with those applied to the original design.

The NRC inspection team reviewed Doosan's measures for using the International Laboratory Accreditation Cooperation (ILAC) accreditation process in lieu of performing commercial-grade surveys for the procurement of calibration services as part of the commercial-grade dedication process. Doosan implements this process as described in the Nuclear Energy Institute document No. 14-05A, "Guidelines for the Use of Accreditation in Lieu of Commercial Grade Surveys for Procurement of Laboratory Calibration and Test Services," Revision 1, dated September 2020, which was recognized for use by the NRC in a safety evaluation (SE) dated November 23, 2020 (Agencywide Documents Access Management System Accession No. ML20322A019).

The NRC inspection team also discussed the design control program with Doosan's management and technical staff. The attachment to this inspection report lists the documents reviewed and personnel interviewed by the NRC inspection team.

b. Observations and Findings

During the review of Doosan's implementation of the ILAC accreditation process, the NRC inspection team noted that it is not being adequately implemented in accordance with the requirements of Revision 1 of NEI 14-05A and the NRC's SE. Specifically, for a sample of the certificates of calibration and laboratory documentation reviewed, the NRC inspection team noted that none of the documents certified that the PO requirements were met. One of the conditions from the NRC's SE states that at receipt inspection, it is

validated that: (1) the laboratory's documentation certifies that the calibration was performed in accordance with the laboratory's 2017 edition of the International Standard Organization (ISO)/International Electrotechnical Commission (IEC) 17025, "General Requirements for the Competence of Testing and Calibration Laboratories," and (2) that the PO requirements were met.

The NRC inspection team determined this issue to be minor because: (1) it is a documentation issue with no impact on the calibration services provided; and (2) the NRC inspection team confirmed the laboratories were accredited to the 2017 edition of ISO/IEC 17025. Doosan initiated CAR No. 2440050 to address this issue.

c. Conclusion

With the exception of the minor issue identified above, the NRC inspection team concluded that Doosan is implementing its design control program in accordance with the regulatory requirements of Criterion III of Appendix B to 10 CFR Part 50, and of 10 CFR 72.146 of Subpart G to 10 CFR Part 72. Based on the limited sample of documents reviewed, the NRC inspection team also determined that Doosan is implementing its policies and procedures associated with the design control program. No findings of significance were identified.

3. Procurement Document Control and Supplier Oversight

a. Inspection Scope

The NRC inspection team reviewed Doosan's policies and implementing procedures that govern the implementation of its procurement document control and supplier oversight programs to verify compliance with the regulatory requirements of Criterion IV, "Procurement Document Control," and Criterion VII, "Control of Purchased Material, Equipment, and Services," of Appendix B to 10 CFR Part 50, and of 10 CFR 72.148, "Procurement document control," and 10 CFR 72.154, "Control of purchased material, equipment, and services," of Subpart G to 10 CFR Part 72. The NRC inspection team reviewed a sample of POs, Doosan's Approved Suppliers List, supplier audit reports, and annual evaluations.

For the review of the sample of POs, the NRC inspection team verified the POs included, as applicable: (1) the scope of work; (2) right of access to the suppliers' facilities; (3) extension of contractual requirements to sub-suppliers; (4) and the applicable technical, regulatory, and quality requirements.

The NRC inspection team also reviewed a sample of audit reports and verified that the audits reports included, as applicable: (1) an audit plan; (2) any findings identified and the associated corrective actions; (3) adequate documented objective evidence of compliance with the applicable requirements; and (4) a documented review by Doosan's responsible management. For the review of the annual evaluations, the NRC inspection team confirmed they included the information required by Doosan's policies and procedures. In addition, the NRC inspection team also verified that the audits were performed in accordance with the established frequency and by qualified lead auditors.

The NRC inspection team also reviewed the training and qualification records of lead auditors and auditors and confirmed that auditing personnel had completed all the

required training and had maintained the applicable qualification and certification in accordance with Doosan's policies and procedures.

The NRC inspection team also discussed the procurement document control and supplier oversight programs with Doosan's management and technical staff. The attachment to this inspection report lists the documents reviewed and personnel interviewed by the NRC inspection team.

b. Observations and Findings

During the review of an audit of a supplier of safety-related machining services, the NRC inspection team noted that Doosan did not include all of the applicable quality assurance (QA) criteria from Appendix B to 10 CFR Part 50 as part of the scope of the audit and as such, it did not adequately verify that the supplier was implementing all of the applicable QA criteria. While a supplier of safety-related machining services is only required to meet the QA criteria applicable to their scope of supply, in this instance, Doosan did not review the implementation of all of the applicable QA criteria for a supplier of safety-related machining services (e.g., corrective action, records, audits).

The NRC inspection team determine this issue to be minor because Doosan verified the rest of the applicable QA criteria, and the machined components are subject to 100% inspection by Doosan upon receipt. Doosan initiated CAR No. 2440055 to address this issue.

c. Conclusion

With the exception of the violation identified in Section 1 of this report, and of the minor issue identified above, the NRC inspection team concluded that Doosan is implementing its procurement document control and supplier oversight programs in accordance with the regulatory requirements of Criterion IV and Criterion VII of Appendix B to 10 CFR Part 50, and of 10 CFR 72.148 and 10 CFR 72.154 of Subpart G to 10 CFR Part 72. Based on the limited sample of documents reviewed, the NRC inspection team also determined that Doosan is implementing its policies and procedures associated with the procurement document control and supplier oversight programs. No findings of significance were identified.

4. Material Traceability

a. Inspection Scope

The NRC inspection team reviewed Doosan's policies and implementing procedures that govern the implementation of its material traceability program to verify compliance with the regulatory requirements of Criterion VIII, "Identification and Control of Materials, Parts, and Components," of Appendix B to 10 CFR Part 50.

The NRC inspection team performed a walkdown of Doosan's forging shop and observed that shop travelers identified the heats for the material, and that all the materials were marked with unique identifiers traceable to procurement documents and Certified Material Test Reports (CMTRs). For the sample of CMTRs reviewed, the NRC inspection team confirmed that the identification markings were: (1) traceable to design and shop drawings, (2) remained legible through the manufacturing process, and that (3)

applied using materials and methods that provided a clear and legible identification and did not adversely affect the function or service life of structure systems or components observed.

The NRC inspection team discussed the material traceability program with Doosan's management and technical staff. The attachment to this inspection report lists the documents reviewed and personnel interviewed by the NRC inspection team.

b. Observation and Findings

No findings of significance were identified.

c. Conclusion

The NRC inspection team concluded that Doosan is implementing its material traceability program in accordance with the regulatory requirements of Criterion VIII of Appendix B to 10 CFR Part 50. Based on the limited sample of documents reviewed, the NRC inspection team also determined that Doosan is implementing its policies and procedures associated with the material traceability program. No findings of significance were identified.

5. Control of Special Processes

a. Inspection Scope

The NRC inspection team reviewed Doosan's policies and implementing procedures that govern the implementation of its control of special processes program to verify compliance with the regulatory requirements of Criterion IX, "Control of Special Processes," of Appendix B to 10 CFR Part 50, and with the applicable requirements of Section III, "Rules for Construction of Nuclear Facility Components," Section V, "Nondestructive Examination," and Section IX, "Welding and Brazing Qualification," of the American Society of Mechanical Engineers (ASME) Boiler and Pressure Vessel (B&PV) Code, and the American Society for Nondestructive Testing (ASNT) SNT-TC-1A, "Personnel Qualification and Certification in Nondestructive Testing."

For manufacturing activities, the NRC inspection team reviewed a sample of shop travelers, work instructions, and the calibration certificates of the measuring equipment. For non-destructive examination (NDE) activities, the NRC inspection team reviewed procedures for magnetic particle testing (MT), liquid penetrant testing (PT), and Ultrasonic Testing (UT), as well as MT, PT, and UT inspector qualifications, and the calibration certificates of the inspection equipment.

Control of Weld Material

The NRC inspection team observed the weld material storage and issue station and verified that weld materials were adequately controlled. Specifically, the NRC inspection team observed that flux covered weld electrodes were either in hermetically sealed containers or kept in baking or portable ovens to control the moisture content within the requirements of the applicable filler metal specification and Section III and Section IX of the ASME B&PV Code. The NRC inspection team also verified that calibrated thermocouples for the baking ovens were used and had valid calibration documentation in accordance with the applicable Doosan procedures.

Nondestructive Examination

The NRC inspection team observed UT on tubesheet forging No. F231890-030 for the Surry Power Station Units 1 and 2 RSGs performed in accordance with Process Specification No. FN231890, "Process Specification for NDE Procedure for Surry 1 & 2 Tubesheet," Revision 7, as documented in traveler No. F23189030.

The NRC inspection verified that the examinations were performed by qualified personnel and qualified procedures in accordance with the applicable requirements of Section III and Section V of the ASME B&PV Code and ASNT SNT-TC-1A. The NRC inspection team also verified that the NDE test reports included all inspection results with the associated inspector and inspection procedures. The NRC inspection team reviewed the Doosan procedures for MT and UT inspections and verified that they were consistent with the applicable ASME B&PV Code requirements.

Heat Treatment

The NRC inspection team observed localized post weld heat treatment of the lifting lugs that were welded on tubesheet forging No. F23189-010 for the Surry Power Station Units 1 and 2 RSGs performed in accordance with heat treat work order No. 238-300 as documented in traveler No. F23189060-2. The NRC inspection team verified that heat treatment and the heat treatment procedure, PTC-NSSS-001, "Heat Treatment Work Procedure Surry 1 & 2," Revision 0, met the applicable requirements of Section III of the ASME B&PV Code and the material specification (ASME Code, Section II, "Ferrous Material Specifications (SA-451-to End)," SA-508, "Specification for Quenched and Tempered Vacuum-Treated Carbon and Alloy Steel Forgings for Pressure Vessels") for quenched and tempered alloy steel. The NRC inspection team also verified that the non-contact pyrometer used to measure the heat treatment temperature was properly calibrated and that the appropriate material traceability controls and heat treatment start time were identified.

The NRC inspection team reviewed heat treatment work instruction No. HF233040, "Surry 1 & 2 Lower Shell Barrel A," Revision 0, used for the lower shell barrel No. F2330400 for the Surry Power Station Units 1 and 2 RSGs met the requirements of Section III of the ASME B&PV Code III and the material specification (ASME B&PV Code, Section II, SA-508) for normalizing heat treatment. The NRC inspection team also verified that the temperature chart recorder and thermocouples used were properly calibrated.

Qualification and Training of Welding and NDE Personnel

The NRC inspection team reviewed the associated welder qualification records and confirmed that the welders had completed the required training and had maintained their qualifications in accordance with Doosan's procedures. The NRC inspection team also verified that the applicable procedure for welder qualification meets the requirements of Section III and Section IX of the ASME B&PV Code.

The NRC inspection team reviewed the NDE inspector qualification records and confirmed they were qualified in accordance with the requirements in ASNT SNT-TC-1A and Doosan's implementing procedures. The NRC inspection team also verified that the applicable implementing procedure for NDE inspector qualification meets the requirements of Sections V of the ASME B&PV Code and ASNT SNT-TC-1A.

The NRC inspection team also discussed the control of special processes program with Doosan's management and technical staff. The attachment to this inspection report lists the documents reviewed and personnel interviewed by the NRC inspection team.

b. Observation and Findings

During a walkdown of the nuclear fabrication shop, the NRC inspection team observed seven forgings machined to specific tolerances for Project No. N22052 for the NuScale Power, LLC. upper reactor pressure vessel. The NRC inspection asked Doosan to provide the controls in place for the storage, cleaning, and preservation of the forgings to prevent damage or deterioration.

Section 13.2.1 of Doosan's Quality Assurance (QA) Manual QM-200 states, in part, that "Each Team and Shop involved with handling, storage, cleaning, packaging, preservation and shipping of all Code items shall prepare process specification, written procedures or work instructions to control these activities while the items are under their responsibility. These documents shall be prepared, reviewed, approved and controlled as described in Section 6.0 by the responsible organization." In addition, Section 5.2.6 of the QA Manual requires work instructions to be prepared for cleaning, handling, and preservation of material.

Doosan stated that it did not have a procedure or inspection instructions for the storage, cleaning, and preservation to prevent damage or deterioration for the forgings. Without objective evidence showing that the forgings had been properly stored and preserved, contaminants can affect the forgings and leave the usability of the forgings as indeterminate.

The NRC inspection team determine this issue to be minor because these forgings are procured as non-safety related. Doosan initiated CAR No. 240054 to address this issue.

c. Conclusion

The NRC inspection team concluded that Doosan is implementing its control of special processes program in accordance with the regulatory requirements of Criterion IX of Appendix B to 10 CFR Part 50. Based on the limited sample of documents reviewed, the NRC inspection team also determined that Doosan is implementing its policies and procedures associated with the control of special processes program. No findings of significance were identified.

6. Test Control

a. Inspection Scope

The NRC inspection team reviewed Doosan's policies and implementing procedures that govern the implementation of its test control program to verify compliance with the regulatory requirements of Criterion XI, "Test Control," of Appendix B to 10 CFR Part 50.

The NRC inspection team reviewed a sample of Doosan's work instructions, including MP-Surry-RSG-001, "Surry 1 & 2 Replacement SG Testing," Revision 2, dated January 4, 2024, for F23189-010 Surry Tubesheet, which included the chemical and mechanical testing of the tubesheet forgings for the RSGs for Surry Power Station Units 1 and 2.

The NRC inspection team selected a sample of four testing reports which included drop weight testing, hardness testing, Charpy V-Notch testing, and chemical analysis testing. These test were performed in accordance with American Society for Testing and Materials (ASTM) A370, "Standard Test Methods and Definitions for Mechanical Testing of Steel Products," which provide the chemical and mechanical verification that the material used meets the required material specification of ASME SA508, Grade 3, Class 2. The NRC inspection team verified that the test reports adequately recorded the test parameters, test results, measuring and test equipment (M&TE) used, test procedure revision and date, test technician, and test results verification.

The NRC inspection team discussed the test control program with Doosan's management and technical staff. The attachment to this inspection report lists the documents reviewed and personnel interviewed by the NRC inspection team.

b. Observations and Findings

No findings of significance were identified.

c. Conclusion

The NRC inspection team concluded that Doosan is implementing its test control program in accordance with the regulatory requirements of Criterion XI of Appendix B to 10 CFR Part 50. Based on the limited sample of documents reviewed, the NRC inspection team also determined that Doosan is implementing its policies and procedures associated with the test control program. No findings of significance were identified.

7. Control of Measuring and Test Equipment

a. Inspection Scope

The NRC inspection team reviewed Doosan's policies and implementing procedures that govern the implementation of its M&TE program to verify compliance with the regulatory requirements of Criterion XII, "Control of Measuring and Test Equipment," of Appendix B to 10 CFR Part 50.

The NRC inspection team verified that the M&TE program ensured that devices used in activities affecting quality were of the proper range, type, and accuracy to verify conformance to established requirements and had the appropriate calibration stickers and current calibration dates, including the calibration due date. In addition, the calibration records reviewed by the NRC inspection team included the as-found or as-left conditions, accuracy required, calibration results, calibration dates, and the due date for recalibration. Furthermore, the NRC inspection team verified that the selected M&TE was calibrated using procedures traceable to known industry standards.

The NRC inspection team also confirmed that when M&TE equipment is found to be out of calibration, Doosan generates a nonconformance report (NCR) in accordance with QCP-1201A, "Control and Calibration of Inspection, Measuring and Test Equipment," Revision 44, July 19, 2023. The NCR will identify items that have been accepted using

this equipment since the last valid calibration date and requires the performance of an extent-of-condition review.

The NRC inspection team also discussed the M&TE program with Doosan's management and technical staff. The attachment to this inspection report lists the documents reviewed and personnel interviewed by the NRC inspection team.

b. Observations and Findings

No findings of significance were identified.

c. Conclusion

The NRC inspection team concluded that Doosan is implementing its M&TE program in accordance with the regulatory requirements of Criterion XII of Appendix B to 10 CFR Part 50. Based on the limited sample of documents reviewed, the NRC inspection team also determined that Doosan is implementing its policies and procedures associated with the M&TE program. No findings of significance were identified.

8. Nonconforming Materials, Parts, or Components and Corrective Action

a. Inspection Scope

The NRC inspection team reviewed Doosan's policies and implementing procedures that govern the implementation of its nonconforming materials, parts, or components and corrective action programs to verify compliance with the regulatory requirements of Criterion XV, "Nonconforming Materials, Parts, or Components," and Criterion XVI, "Corrective Action," of Appendix B to 10 CFR Part 50, and 10 CFR 72.170, "Nonconforming materials, parts, or components," and 10 CFR 72.172, "Corrective action," of Subpart G to 10 CFR Part 72. The NRC inspection team reviewed Doosan's NCRs and CAR logs and selected a sample of NCRs and CARs to verify that Doosan had implemented an adequate program to ensure that nonconforming items, significant conditions adverse to quality, and conditions adverse to quality were promptly identified and corrected. The NRC inspection team verified that Doosan's processes and procedures provided for the identification, documentation, segregation, evaluation, and disposition of nonconforming items. Nonconformances can be dispositioned as "Use-As-Is," "Repair," "Rework," or "Reject."

The NRC inspection team reviewed a sample of NCRs and confirmed that Doosan: (1) dispositioned the NCRs in accordance with Doosan's applicable policies and procedures; (2) documented an appropriate technical justification for various dispositions; and (3) took adequate corrective action with regards to the nonconforming items. The NRC inspection team also verified that Doosan's NCRs provide a link to the 10 CFR Part 21 program.

The NRC inspection team also reviewed a sample of CARs and confirmed that the CARs contained, as applicable: (1) adequate documentation and description of conditions adverse to quality; (2) an appropriate analysis of the cause of these conditions and the corrective actions taken to prevent recurrence; (3) direction for review and approval by the responsible authority; (4) a description of the current status of the

corrective actions; and (5) the actions taken to verify timely and effective implementation of the corrective actions. The NRC inspection team also verified that Doosan's CARs provide a link to the 10 CFR Part 21 program.

In addition to NCRs and CARs, Doosan issues Audit Deficiency Reports (ADRs) to address deficiencies identified during Doosan's internal audits, audits of Doosan performed by its customers, and audits performed by Doosan of its suppliers. ADRs function similarly as CARs. The NRC inspection team reviewed a sample of ADRs and confirmed that the ADRs all the required information in accordance with Doosan's policies and procedures. The NRC inspection team also verified that Doosan's ADRs provide a link to the 10 CFR Part 21 program.

The NRC inspection team also discussed the nonconforming materials, parts, or components and corrective action programs with Doosan's management and technical staff. The attachment to this inspection report lists the documents reviewed and personnel interviewed by the NRC inspection team.

b. Observations and Findings

No findings of significance were identified.

c. Conclusion

The NRC inspection team concluded that Doosan is implementing its nonconforming materials, parts, or components and corrective action programs in accordance with the regulatory requirements of Criterion XV and Criterion XVI of Appendix B to 10 CFR Part 50, and of 10 CFR 72.170 and 10 CFR 72.172 of Subpart G to 10 CFR Part 72. Based on the limited sample of documents reviewed, the NRC inspection team also determined that Doosan is implementing its policies and procedures associated with the control of nonconforming materials, parts, or components and corrective action programs. No findings of significance were identified.

9. Internal Audits

a. Inspection Scope

The NRC inspection team reviewed Doosan's policies and implementing procedures that govern the implementation of its internal audits program to verify compliance with the regulatory requirements of Criterion XVIII, "Audits," of Appendix B to 10 CFR Part 50 and 10 CFR 72.176, "Audits," of Subpart G to 10 CFR Part 72.

For the sample of internal audits reviewed, the NRC inspection team verified that the audit reports included: (1) an audit plan; (2) audit results; (3) adequately documented objective evidence with the applicable requirements; and (4) a review by Doosan's responsible management. The NRC inspection team verified that the internal audits were performed by qualified auditors who were not auditing their own work and that the internal audits were performed using the appropriate checklists. The NRC inspection team also verified that Doosan adequately initiated corrective actions for any findings identified during the internal audits.

The NRC inspection team also discussed the internal audits program with Doosan's management and technical staff. The attachment to this inspection report lists the documents reviewed and personnel interviewed by the NRC inspection team.

b. Observations and Findings

No findings of significance were identified.

c. Conclusion

The NRC inspection team concluded that Doosan is implementing its internal audits program in accordance with the regulatory requirements of Criterion XVIII of Appendix B to 10 CFR Part 50 and of 10 CFR 72.176 of Subpart G to 10 CFR Part 72. Based on the limited sample of documents reviewed, the NRC inspection team also determined that Doosan is implementing its policies and procedures associated with the internal audits program. No findings of significance were identified.

10. Entrance and Exit Meetings

On May 20, 2024, the NRC inspection team discussed the scope of the inspection with Mr. Dongsoo Lee, Senior Vice President, Nuclear Design/Manufacturing, and other members of Doosan's management and technical staff. On May 24, 2024, the NRC inspection team presented the inspection results and observations during an exit meeting with Mr. Kyoungsik Choi, Vice President, Quality Assurance, and other members of Doosan's management and technical staff. The attachment to this report lists the attendees of the entrance and exit meetings, as well as those individuals whom the NRC inspection team interviewed.

ATTACHMENT

1. ENTRANCE/EXIT MEETING ATTENDEES

Name	Title	Affiliation	Entrance	Exit	Interviewed
Dongsoo Lee	Senior Vice President Nuclear Design & Manufacturing	Doosan Enerbility, Ltd. (Doosan)	X		
Kyoungsik Choi	Quality Assurance (QA) Vice President	Doosan	X	X	
Yongkyu Kim	Nuclear Design Vice President	Doosan	X	X	
Namsoo Choi	Casting & Forging Production Vice President	Doosan	X	X	
Wonman Lee	Nuclear Quality Control (QC) Senior Director	Doosan	X	X	
Jeongsun Kim	QA Team General Manager	Doosan	X	X	X
Jeongsuk Lee	Cask PM General Manager	Doosan	X	X	
Jinmin Wang	Nuclear Manufacturing Engineering Team General Manager	Doosan	X	X	
Jihun Kim	Nuclear QC Team 2 General Manager	Doosan	X	X	
Woodong Lee	Casting & Forging Materials Engineering General Manager	Doosan	X		
Kisu Kwon	Casting & Forging QC Team General Manager	Doosan	X	X	
Kyeongseon Lim	QA Team Senior Engineer	Doosan	X	X	X

Name	Title	Affiliation	Entrance	Exit	Interviewed
Laehyuck Jeong	Nuclear Production Strategy Team Senior Engineer	Doosan	X		
Byeol Baek	Casting & Forging Production Control Team Senior Engineer	Doosan	X		
Jeonghyun Lee	Casting & Forging Quality Control Team Senior Engineer	Doosan	X	X	X
Jaewoong Kim	QA Team Senior Engineer	Doosan	X	X	X
Hyunchang Kim	Nuclear QC Team 2 SMR Inspection Part Leader	Doosan	X	X	X
Hyunjun Jeong	Steam Generator Design Team Senior Engineer	Doosan	X	X	X
Yoseb Cha	Cask Design Team General Manager	Doosan		X	
Dongjin Goo	Cask Design Team Senior Engineer	Doosan		X	
Minkun Bae	Casting & Forging Production Control Team General Manager	Doosan		X	
Hwanhong Kim	Senior Manager	Doosan			X
Seungwook Kim	Senior Engineer	Doosan			X
Chulmin Shin	Senior Engineer	Doosan			X
Heonjong Lee	Senior Engineer	Doosan			X
Kiseong Nam	Senior Engineer	Doosan			X
Youngchang Song	Engineer	Doosan			X
Jae Nyeong Choi	Engineer	Doosan			X
Hoyong Kim	Calibration Engineer	Doosan			X

Name	Title	Affiliation	Entrance	Exit	Interviewed
Samjin Kim	Calibration Engineer	Doosan			X
Mingi Choi	Calibration Engineer	Doosan			X
Y.C. Jeong	Heat Treatment Operator	Doosan			X
S. W. Kang	Heat Treatment Operator	Doosan			X
J.H. Yun	Welder	Doosan			X
C.E. Kim	Welder	Doosan			X
Jeong Hwan Lee	Nondestructive Examination (NDE) Inspector	Doosan			X
Heon Seong Jang	NDE Inspector	Doosan			X
D. M. Cha	NDE Engineer	Doosan			X
N. K. Kang	Mechanical Test Engineer	Doosan			X
Jinguk Jang	Resident Inspector	Westinghouse Electric Company	X	X	
Tim O'Brian	Inspector	Dominion Generation		X	
J.G. Yeon	QA Engineer	Koran Institute of Nuclear Safety (KINS)	X	X	
Myunghyun Yoon	QA Senior Researcher	KINS	X	X	
Yamir Diaz-Castillo	Inspection Team Leader	Nuclear Regulatory Commission (NRC)	X	X	
John Honcharik	Inspector	NRC	X	X	
Marlone Davis	Inspector	NRC	X	X	

2. INSPECTION PROCEDURES USED

- Inspection Procedure (IP) 36100, "Inspection of 10 CFR Part 21 and Programs for Reporting Defects and Noncompliance," dated February 10, 2023

- IP 43002, "Routine Inspections of Nuclear Vendors," dated February 10, 2023
- IP 43004, "Inspection of Commercial-Grade Dedication Programs," dated February 10, 2023
- IP 60852, "ISFSI Component Fabrication by Outside Fabricators," dated August 17, 2023

3. LIST OF ITEMS OPENED, CLOSED, AND DISCUSSED

Item Number	Status	Type	Description
999021373/2024-201-01	OPENED	Notice of Violation (NOV)	10 CFR Part 21

4. DOCUMENTS REVIEWED

Policies and Procedures

- QM-200, "Quality Assurance Manual for ASME III and KEPIC-MN & SN Construction including Furnishing Material and Service and NBIC Repair/Alternation," Revision 65, dated February 8, 2024
- Quality Assurance Procedure No. PQAP-1602, "Reporting of 'Defects' and 'Failures to Comply Pursuant to 10CFR21," Revision 13, dated May 10, 2024
- Quality Control Procedure (QCP) - 0204, "NDE Personnel Certification Program," Revision 29, dated March 18, 2024
- QCP-0205, "Procedure for Qualification of Audit Personnel," Revision 18, dated May 13, 2022
- QCP-0206, "Qualification and Certification of Welding and Brazing Personnel," Revision 27, dated March 15, 2024
- QCP-401, "Procurement Documents Control Procedure," Revision 36, dated May 2, 2024
- QCP-0701, "Vendor Evaluation for Nuclear Project," Revision 44, dated March 20, 2024
- QCP-0702, "Control of Subcontracted Services," Revision 22, dated March 20, 2024
- QCP-0902, "Qualification and Control of Welding and Brazing Procedure Specification," Revision 26, dated March 15, 2024
- QCP-1201B, "Control and Calibration of Production Equipment, Revision 22, dated July 7, 2022
- QCP-1201A, "Control and Calibration of Inspection, Measuring and Test Equipment," Revision 44, dated July 19, 2023

- QCP-1202, "Control and Calibration of Non-Destructive Examination Equipment," Revision 13, dated March 31, 2023
- QCP-1301, "Control of Handling, Storage, Preservation, Packaging and Shipping," Revision 18, dated November 23, 2023
- QCP-1302, "Control of Materials," Revision 23, dated January 30, 2023
- QCP-1801, "Nuclear Project QA Survey/Audits," Revision 34, dated May 30, 2022
- QCP-1803, "Quality Surveillance," Revision 10, dated May 26, 2022
- QCP-1804, "Management Review and QA Program Effectiveness Evaluation," Revision 10, dated March 14, 2024
- QCP-2203, "Preparation, Certification and Control of ASME Code/KEPIC Data Reports," Revision 10, dated May 27, 2023
- QCP No. CQCP-0201N, "Indoctrination & Training Procedure," Revision 16, dated February 19, 2024
- QCP No. CQCP-0401N, "Control of Procurement Documents," Revision 14, dated June 14, 2022
- QCP No. CQCP-0702N, "Control of Subcontracted Services," Revision 11, dated June 8, 2022
- QCP No. QCP-1201A, "Control of Calibration of Measuring and Test Equipment," Revision 38, dated January 7, 2022
- QCP No. QCP-1301, "Control of Handling, Storage, Preservation, Packaging and Shipping," Revision 18, dated November 23, 2023
- QCP No. CQCP-1501N, "Control and Correction of Nonconforming Items and Activities," revision 18, dated June 8, 2022
- QCP N0. CQCP-1602N, "Control of Corrective Actions," Revision 8, dated June 8, 2022
- Nuclear Quality Control Procedure (NQCP) No. NQCP-0201, "Indoctrination & Training Procedure," Revision 19, dated October 25, 2023
- NQCP-0203, "Qualification of Inspection and Test Personnel," Revision 23, dated April 28, 2022
- NQCP-0207, "Qualification of Heat Treatment Operators," Revision 10, dated March 18, 2024
- NQCP-705N, "Control of Welding Material," Revision 30, dated March 18, 2024

- NQCP-0706, "Commercial Grate Item Dedication Procedure," Revision 12, dated April 9, 2024
- NQCP-0801, "Control of Identification and Traceability of Items and Materials," Revision 9, dated July 6, 2022
- NQCP-0901, "Control of Production Processes," Revision 28, dated March 28, 2023
- NQCP-0904, "Qualification and Control of Procedures for forming Process, Revision 8, dated March 16, 2023
- NQCP-0905, "Control of Heat Treatment Operations, Revision 13, dated March 18, 2024
- NQCP No. NCQP-1501, "Control and Correction of Nonconforming Items and Activities," Revision 33, dated July 18, 2023
- NQCP No. NQCP-1602, "Control of Corrective Actions," Revision 22, dated July 14, 2023
- NQCP No. NQCP-1607, "Site NCR Control Procedure," Revision 18, dated March 9, 2023
- Document No. U037-QAR-01, "Quality Assurance Requirements for ASME Code Class 1, 2 and 3 Materials," Revision A (no date provided)
- Document No. N14002-111PS-0020, "Preservation & Maintenance Plan for SQN Units 1&2 and WBN Units 1 &2 Replacement Reactor Vessel Closure Head Forging," Revision 3, dated March 18, 2024
- Work Instruction No. HF233040, "Surry 1 & 2 Lower Shell Barrel A," Revision 0
- Work Instruction No. PS-55454, "Calibration of Emission Vacuum Spectrometer," Revision 3, dated January 31, 2005
- Work Instruction No. PTC-NSSS-001, "Heat Treatment Work Procedure Surry 1 & 2," Revision 0
- Work Instruction No. FN23189-030, "Process Specification for NDE Procedure for Surry 1 & 2 Tubesheet," Revision 7
- Work instruction MP-Surry-RSG-001, "Surry 1 & 2 Replacement SG Testing," Revision 2, dated January 4, 2024
- Drawing No. UT231890, "Rough Machining for UT (Tubesheet)," Revision 1, dated October 24, 2023
- Drawing No. QH231890, "Rough Machining for QHT (Tubesheet)," Revision 1, dated October 23, 2024

- Traveler No. F23189030, Surry Power Station Units 1 and 2 for tubesheet forging No. F231890-030
- Traveler No. F23189060 for Surry Power Station Units 1 and 2 tubesheet forging No. F23189-060.
- Traveler No. F23189020, Surry Power Station Units 1 and 2 for tubesheet forging No. F231890-020
- Traveler No. 101590850, Watts Bar Nuclear Power Plant Unit 2 Reactor Vessel Closure Head Preservation, dated May 3, 2018
- Traveler No. 101920517, Watts Bar Nuclear Power Plant Unit 2 Reactor Vessel Closure Head Preservation, dated April 11, 2024
- UT231891, "Rough Machining for UT," Revision 1, dated October 24, 2023
- VPAM-SG000-DW-CE-000001, "Surry Units 1 and 2 Model 53F Replacement Steam Generator Tubesheet Forging," Revision 2, dated March 21, 2024
- VPAM-SG000-DW-CE-000006, "Surry Units 1 and 2 Model 53F Replacement Steam Generator Transition Cone Forging," Revision 1, dated September 15, 2023
- Drawing No. 5746-90-001, "Forging Archive Samples," Revision B, dated October 13, 2023
- Drawing No. 30097-061, "Structure, Weldment, Concrete Cask, MAGNASTOR, TMI, Revision 0, dated October 24, 2021
- Drawing No. 30097-062, "Structure, Weldment, Concrete Cask, MAGNASTOR, TMI, Revision 0, dated October 24, 2021
- Drawing No. 30097-064, "Upper Segment Assembly, Concrete Cask, MAGNASTOR, TMI, Revision 0, dated November 2, 2021
- Drawing No. 30097-090, "Loaded Concrete Cask Assembly," Revision 0, dated October 24, 2021
- Drawing No. N21055-333MD-2300, "Top Support and Liner Shell Assembly of Liner Weldment," Revision 2, dated December 12, 2022
- Drawing No. N21055-333MD-2500, "Final Machining of Liner Weldment," Revision 3, dated January 26, 2023
- Drawing No. N21055-333MD-3500, "Rail Assembly of Bottom Weldment," Revision 1, dated December 1, 2022
- Drawing No. N21055-333MD-3700, "Final Machining of Bottom Weldment," Revision 1, dated December 9, 2022

- Drawing No. N21055-333MD-4500, "Lift Lug Support Plate, Lifting Tab and Shielding Sector Welding of Upper Segment Weldment," Revision 4, dated October 18, 2023
- Drawing No. N21055-333MD-5100, "Outlet and Inlet Screen Assembly," dated January 12, 2022
- Report of Ultrasonic Examination Primary Side Scan U240507-020-001, Tubesheet (MFOR030088) Forging F23189-030, dated May 10, 2024
- Report of Ultrasonic Examination Secondary Side Scan U240517-007-001, Tubesheet (MFOR030088) Forging F23189-030, dated May 21, 2024
- Report of Ultrasonic Examination Primary Side Scan U240501-001-001, Tubesheet (MFOR030088) Forging F23189-020, dated May 7, 2024
- Report of Ultrasonic Examination Secondary Side Scan U240509-014-001, Tubesheet (MFOR030088) Forging F23189-020, dated May 17, 2024
- Report of Ultrasonic Examination Primary Side Scan U240426-009-001, Tubesheet (MFOR030088) Forging F23189-010, dated April 30, 2024
- Report of Ultrasonic Examination Secondary Side Scan U240503-005-001, Tubesheet (MFOR030088) Forging F23189-010, dated May 8, 2024
- Report Of Magnetic Particle Examination No. M240430-025-001, Tubesheet F23189-010, May 2, 2024, Primary Side
- Report Of Magnetic Particle Examination No. M240503-003-001, Tubesheet F23189-010, May 6, 2024, Secondary Side surface
- U037-QAR-01, "Quality Assurance Requirements for ASME Code Class 1,2 and 3 Materials," Revision C, dated April 11, 2024
- VPAM-SG000-SP-CM-000001, "SA-508 Grade 3 Class 2, Forgings (Section III-NB)," Revision 1, dated July 25, 2023
- MP-Surry-RSG-1001, "TubeSheet," Revision 3, dated April 17, 2024
- N23029-PS, "General Quality Requirements for Surry 12 RSG," Revision 1, dated January 10, 2024
- 71160-S-06, "Procurement Fabrication Specification, MAGNASTOR Concrete Cask Steel Weldments and Components," Revision 2, dated July 28, 2022
- 71160-S-06, "Procurement Fabrication Specification, MAGNASTOR"
- NPS-11105F, "General Quality Requirements of material for Non-ASME Section III and/or Non-KEPIC MN&SN (for foreign company)," Revision 11, dated June 30, 2022

- N20007-000PS-1001, "Temporary Packing and Storage Plan for TMI TSC," Revision 0, dated May 10, 2021
- N20007-000PS-0001, "Cleaning Procedure for NAC TMI TSC," Revision 2, dated July 7, 2020

Calibration, Inspection and Test Control

- Dimensional Inspection Record No. F23189010-6151, dated April 22, 2024
- Calibration Requisition and Receipt for a survey meter, gage ID No. 08D4M00170023, dated April 11, 2024
- Calibration Certificate No. 24-SM-0675 for a survey meter, gage ID No. 08D4M00170023, dated April 22, 2024
- Dedication Report for Commercial Calibration Service No. Z240419024 for Calibration Certificate No. 24-SM-0675, dated April 25, 2024
- Calibration Requisition and Receipt for a digital vernier caliper, gage ID No. 08A3G00120119, dated April 3, 2024
- Calibration Certificate No. K240183A-106-0629 for a digital vernier caliper, gage ID No. 08A3G00120119, dated April 3, 2023
- Dedication Report for Commercial Calibration Service No. C240403040 for Calibration Certificate No. K240183A-106-0629, dated April 3, 2024
- Calibration Requisition and Receipt for a dew point meter, gage ID No. 08D8U00400007, dated March 3, 2013
- Calibration Certificate No. C0260BD00314 for a dew point meter, gage ID No. 08D8U00400007, dated March 13, 2023
- Calibration Certificate No. Z230622037, for ID No. 08F1M0011-0001, drop weight testing machine, calibration due date June 22, 2024
- Calibration Certificate No. Z2400217502-043, for ID No. STD04449, non-contact pyrometer, calibration due date January 10, 2025
- Calibration Certificate No. 10073-4210, for a gage block comparator, calibration due date May 11, 2024, Recalibrated by Korean Research Institute of Science and Standards (KRISS) on May 16, 2024 (with no use from May 11 to May 16, 2024)
- Calibration Certificate No. 21-0215-1 for a flux oven, calibration due date July 6, 2024
- Calibration Certificate No. 42-469 for probe No. 121439504, calibration due date July 2, 2024
- Calibration Certificate 42-718 for probe No. 121439103, due June 19, 2024

- Calibration Certificate No. 13-0263 for temperature chart recorder No. 437112, calibration due date June 13, 2024
- Calibration Certificate No. 08F5C0092-0246 for a thermocouple, calibration due date July 17, 2024
- Calibration Certificate No. 08F5C0092-0132 for a thermocouple, calibration due date July 13, 2024
- Calibration Certificate No. 08F5C0383-0038 for a thermocouple, calibration due date July 10, 2024
- Calibration Certificate No. 08F5C0383-0039 for a thermocouple, calibration due date July 10, 2024
- Calibration Certificate No. 08F5C0383-0042 for a thermocouple, calibration due date July 6, 2024
- Calibration Certificate No. 08F5C0383-0043 for a thermocouple, calibration due date July 6, 2024
- Calibration Certificate No. 08F5C0383-0046 for a thermocouple, calibration due date July 10, 2024
- Calibration Certificate No. 08F5C1016-0044 for a thermocouple, calibration due date July 16, 2024
- Calibration Certificate No. 08F5C1016-0045 for a thermocouple, calibration due date July 16, 2024
- Calibration Certificate No. 08F5C1016-0046 for a thermocouple, calibration due date July 13, 2024
- Calibration Certificate No. 08F5C1016-0044 for a thermocouple, calibration due date July 16, 2024
- Dedication Report for Commercial Calibration Service No. for Calibration Certificate No. C0260BD00314, dated March 19, 2023
- Certified Material Test Report (CMTR) No. MCF23189-030, Surry #1,2 RSG Forging and AVB, Tubesheet SA508 GR.3 CL.2 forging No. 23189-030
- CMTR No. MCF23189-020, Surry #1,2 RSG Forging and AVB, Tubesheet SA508 GR.3 CL.2 forging No. 23189-020
- Drop Weight Test Report IP-24-04-541 for F23189-010 Surry Tubesheet, dated April 15, 2024

- Test Report Machining & Test Request 2024-03-29-16 for Surry Tubesheet F23189-010, dated April 12, 2024
- Test Report Hardness Test Report IP-24-04-540 for F23189-010 Surry Tubesheet, dated April 15, 2024
- Test Report Impact Charpy V-Notch Report IP-24-04-539 for F23189-010 dated April 15, 2024
- Test Report Drop Weight Test Report IP-24-04-541 for F23189-010 Surry Tubesheet, dated April 15, 2024
- Chemical Analysis Test Report IC-24-05-0190 for F23189-010 Surry Tubesheet, dated May 2, 2024

Purchase Orders, Audit Reports, and Annual Evaluations

- Doosan's Approved Vendors List
- NuScale Task Order No. CO-1218-63820, Revision 0, dated December 22, 2022
- SW-123109, "NuScale Statement of Work for Upper RPV Long Lead Materials," Revision 5, dated October 31, 2023
- Document No. N22052-110PS-00508, "Material Purchase Specification for SA-508-Grade 3 Class 2 Forgings (To Be Used for Upper RPV) for NuScale Project," Revision 5, dated December 14, 2023
- Purchase Order (PO) No. 2023020550 for machining services, Revision 3, dated February 22, 2024
- PO No. 2024004416 for machining services, Revision 0, dated March 20, 2024
- PO No. 2023008631 for Nickel-Alloy steel seamless tubes, Revision 4, dated April 11, 2024
- PO No. 2023014081 for antivibration bars, Revision 3, dated September 12, 2023
- PO No. 2016007410 for ASTM A36 carbon steel plates, Revision 1 April 21, 2016
- PO No. 2021020167 for ASTM A36 carbon steel plates, Revision 2, dated December 21, 2012
- PO No. 2016022050 for machining services, Revision 0, dated November 3, 2016
- PO No. 2018015024 for hex nuts, Revision 1, dated August 13, 2018
- PO No. 846054 for transportable storage canisters, Revision 2, dated January 24, 2020
- PO No. 846352 for VCC Liners, Revision 3, dated December 5, 2023

- PO No. 202300163 for forgings, Revision 8, dated May 9, 2024
- Task Order Revision 8 to NuScale Contract No. CO-1218-63820, Statement of Work No. SW-123109, Revision 5, dated April 26, 2024, task end date December 31, 2025
- PO No. 4500634344, "Watts Bar 2 Reactor Closure Head and Replacement Steam Generators," dated June 30, 2018
- Purchase Specification (PS) No. CF-MPS-001, "General Requirements (for foreign company)," Revision 17, dated January 20, 2023
- PS No. PS-11102N, "General Quality Requirements for NuScale Project," Revision 2, dated December 13, 2023
- PS No. NPS-11105F, "General Quality Requirements of Material for Non-ASME Section III and/or Non-KEPIC MN&SN (for Foreign Company), Revision 10, dated April 20, 2020
- PS No. NPS-11102-NS, "ASME Section III KEPIC MN/SN," Revision 4, dated December 14, 2015
- PS No. PS-11102WB, "General Purchase Requirements for Watts Bar Unit 2 RSG Project," Revision 0, dated March 26, 204
- PS No. N23029-PS, "General Quality Requirements for Surry 12 RSG," Revision 1 (no date provided)
- PS No. N23029-160PS-0001, "Purchase Specification for Anti-Vibration Bar Assembly for Surry 1 & 2 RSG," Revision 1 (no date provided)
- Survey/Audit Plan No. VSP-2401-02, dated January 18, 2024
- Survey/Audit Plan No. VSP-2202-04, dated February 11, 2022
- Survey/Audit Plan No. VSP-2304-05, dated April 3, 2023
- Survey/Audit Plan No. VSP-2303-05, dated March 15, 2023
- Survey/Audit Plan No. VAP-2304-04, dated April 3, 2023
- Survey/Audit Plan No. VSP-2015-05, dated May 26, 2021
- Survey/Audit Plan No. VAP-2309-09, dated September 25, 2023
- Vendor Survey/Audit Report No. VSR-2401-02, dated February 19, 2024
- Vendor Survey/Audit Report No. VSR-2202-04, dated March 10, 2022
- Vendor Survey/Audit Report No. VSR-2304-05, dated April 3, 2023
- Vendor Survey/Audit Report No. VSR-2323-05, dated March 15, 2023

- Vendor Survey/Audit Report No. VSR-2304-04, dated April 3, 2023
- Vendor Survey/Audit Report No. VSR-2015-05, dated June 10, 2021
- Vendor Survey/Audit Report No. VAR-2309-09, dated October 26, 2023
- Supplier Audit/Survey Checklist No. VCSL-2401-02, dated January 19, 2024
- Supplier Audit/Survey Checklist No. VCSL-2202-04, dated February 14, 2022
- Supplier Audit/Survey Checklist No. VCSL-2304-05, audit/survey date April 25-26, 2023
- Supplier Audit/Survey Checklist No. VCSL-2305-05, audit/survey date March 21-23, 2023
- Supplier Audit/Survey Checklist No. VCSL-2304-04, audit/survey date April 20-21, 2023
- Supplier Audit/Survey Checklist No. VCSL-2025-05, dated May 27, 2021
- Supplier Audit/Survey Checklist No. VACL-2309-09, audit/survey date September 26, 2023
- Annual Vendor Evaluation Report No. VER-2301-01, dated January 5, 2023
- Annual Vendor Evaluation Report No. VER-2403-05, dated March 26, 2024
- Annual Vendor Evaluation Report No. VER-2405-07, dated May 20, 2024
- Annual Vendor Evaluation Report No. VER-2306-03, dated June 6, 2023

Nonconformance Reports

- 20100345, 21100317, 23100254, 23100287, and 24100068.

Corrective Action Reports

- 220161, 230052, 240037, and 240036

Survey/Audit Deficiency Reports

- Audit Deficiency Report (ADR) Nos. 200066, 200169, 200172, 200178, 210071, 210072, 210086, 220001, 220065, 220066, 220143, 220145, 220147, 230104, 230105, 230106, 230107, 230073, 230074, 230075, 230076, 230077, 230078, 230079, 230080, 230095, 230096, 230284, and 240012.

Corrective Action Requests Opened During the NRC Inspection

- 240050, 240051, 240052, 240053, 240054, and 240055.

Internal Audits

- Internal Audit Reports, Plans, Audit Checklists for 2018, 2019, 2021, 2022, and 2023
- Management Review Reports for 2019, 2021, 2022, and 2023

Training and Qualification

- Lead auditor and auditor qualification records for Mr. Bum Yong Kim, Mr. Minkyu Choi, Mr. Sukkwan Kim, Mr. B. S. Seo, Mr. S. P. Seol, and Mr. J. S. Kim.

Miscellaneous

- Nuclear Business Group Quality Trend Analysis 2023